

Decentralized Labor Systems: Rural Connect

**Solving the
Unorganized Labor
Gap in Niche
Agricultural and
Household Services**

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**RURAL
CONNECT**

Bridging Communities, Enabling Livelihoods.

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INTRODUCTION

In rapidly urbanizing and semi-urban landscapes, a significant structural gap exists between households requiring specialized manual labour and the unorganized workforce capable of providing it. Traditional home-service applications heavily prioritize standardized, white-collar, or highly commercialized urban amenities. This leaves highly specialized, regional, and seasonal skill sets—such as traditional masonry (*Rajmistry*), traditional construction, and seasonal agricultural climbing (mango and coconut harvesting)—completely fragmented and digitally excluded.

Rural Connect is an innovative, dual-sided hyperlocal marketplace designed specifically to bridge this gap. By moving away from rigid corporate pricing structures, the platform introduces an **Open Marketplace Pricing Model**. This model empowers independent, skilled manual labourers to set their own rates, choose their own hours, and directly manage their client relationships.

By implementing an **Open Marketplace Pricing Model**, the platform empowers independent workers to set their own rates, driving micro-entrepreneurship while providing consumers with transparent, competitive, and verified local talent options. This document details the strategic product vision, data architecture (SQL), and analytical frameworks (Power BI) necessary to launch, scale, and sustain the platform

Ultimately, Rural Connect shifts the paradigm of casual labour from vulnerable daily-wage work to structured micro-entrepreneurship. By combining a low-friction user experience with a localized, geo-fenced data architecture, Rural connect provides consumers with a transparent, verified pool of local talent while building a safer, predictable, and sustainable digital economy for the unorganized sector.

PROBLEM STATEMENT & SOLUTION

The unorganized labour sector in India accounts for over 80% of the total workforce, yet it remains completely fragmented. Current market dynamics suffer from three critical structural flaws:

1. **The Invisibility of Niche Skills:** Finding a local *Rajmistry* for a small brickwork patch or a certified climber to clear a coconut tree relies entirely on inefficient word-of-mouth networks.
2. **Asymmetric and Arbitrary Pricing:** Because there are no standardized price discovery mechanisms, customers face random price gouging, while workers face exploitation or irregular wage structures.
3. **The Trust Deficit:** Households are hesitant to let unverified labourers onto their private property due to safety concerns, while labourers lack insurance coverage against high-risk manual tasks.

The Solution:

Rural Connect digitizes this ecosystem. It serves as a localized matching engine that introduces digital profiling, verified identity matching, transparent data-driven user reviews, and an autonomous pricing model that brings structural safety and efficiency to the unorganized marketplace.

COMPETITIVE ADVANTAGE

To establish a distinct market position, TaskNeighbor explicitly targets sectors completely ignored by standard corporate service apps:

Core Dimension	Urban Company	Rural Connect
Primary Target Market	Standardized corporate home repairs (AC servicing, salon, appliance fixing).	Niche, highly specialized manual, agricultural, and civil labor.
Agricultural Integration	Non-existent. No capabilities in harvesting or land management.	Deep Integration: Specialized fruit pluckers (Mango/Coconut) available on-demand.
Civil & Heavy Structural Work	Limited to minor handyman/plumbing patches.	Full Competency: Dedicated access to local <i>Rajmistry</i> assets for structural masonry.
Pricing Strategy	Corporate-fixed algorithmically controlled rigid pricing.	Open Marketplace: Autonomous worker-driven bidding and hourly-rate setting.
Geographic Scope	Top-tier dense metro areas only.	Hyperlocal Outskirts: Built for urban fringes, tier-2 cities, and semi-urban hubs.

PRODUCT REQUIREMENT DOCUMENTS

1. USER STORIES (THE CUSTOMER & PROVIDER JOURNEY)

To establish a clear link between market needs and the system features, the platform requirements are derived from real-world User Stories. These stories reflect the operational needs of both stakeholders in the marketplace.

1.1 Customer User Stories

- * As a Customer, I want to see the approximate height of a tree or building tier before booking a service, so that I can accurately communicate the scale of the work and avoid post-arrival disputes.
- * As a Customer, I want to view a real-time rolling average market rate for the specific service tier I choose, so that I can evaluate whether a worker's custom fee is fair.
- * As a Customer, I want to choose a specific worker myself from a transparent roster based on their rating, expertise, and status, so that I maintain control over who enters my property.
- * As a Cash-on-Delivery (COD) Customer, I want to verify the completed work before making a physical cash payment, using an OTP handshake to secure the transaction close out.

1.2 Worker User Stories

- * As a Worker, I want to input and control my own digital rate card, so that I can price my labor according to my experience and specialized skill sets.
- * As a Worker, I want to see explicit job parameters (like tree height or room counts) alongside customer location before I begin traveling, so that I can bring the exact safety gear or tools required.
- * As a Worker, I want an emergency cancel option (Option B) that alerts company dispatch in case of an on-route accident, so that my platform performance metrics are protected during genuine crises.

2. FUNCTIONAL REQUIREMENTS MATRIX

Functional Requirements (FR) define the specific capabilities, inputs, and logical rules that the Rural Connect system must execute to satisfy the user stories above.

2.1 User Authentication & Profile Management

- * FR-1.1: The system must provide separate onboarding and profile flows for "User" and "Worker" roles.
- * FR-1.2: The system must capture and store the worker's name, mobile number, and verified service expertise during registration.

- * FR-1.3: The system must enforce a document upload block for workers, holding account status as "Pending" until national identity verification is approved.

2.2 Core Search & Dynamic Tiering Engine

- * FR-2.1: The system must capture the customer's current GPS location or manual location entry to filter localized workers within a set operational radius.
- * FR-2.2: When a specialized service (Fruit Plucking, Painting, Decoration) is selected, the system must force a dynamic scoping parameter choice (e.g., Tree Height Selector, House Structure, Number of Rooms).
- * FR-2.3: The system must calculate and display a rolling local average rate specifically matching the selected service category and tier.
- * FR-2.4: The system must display a filtered roster of workers matching the service criteria, showing each worker's custom hourly fee, average rating, expertise alignment, and real-time status (Online / On-Job).

2.3 Cart & Financial Checkout Processing

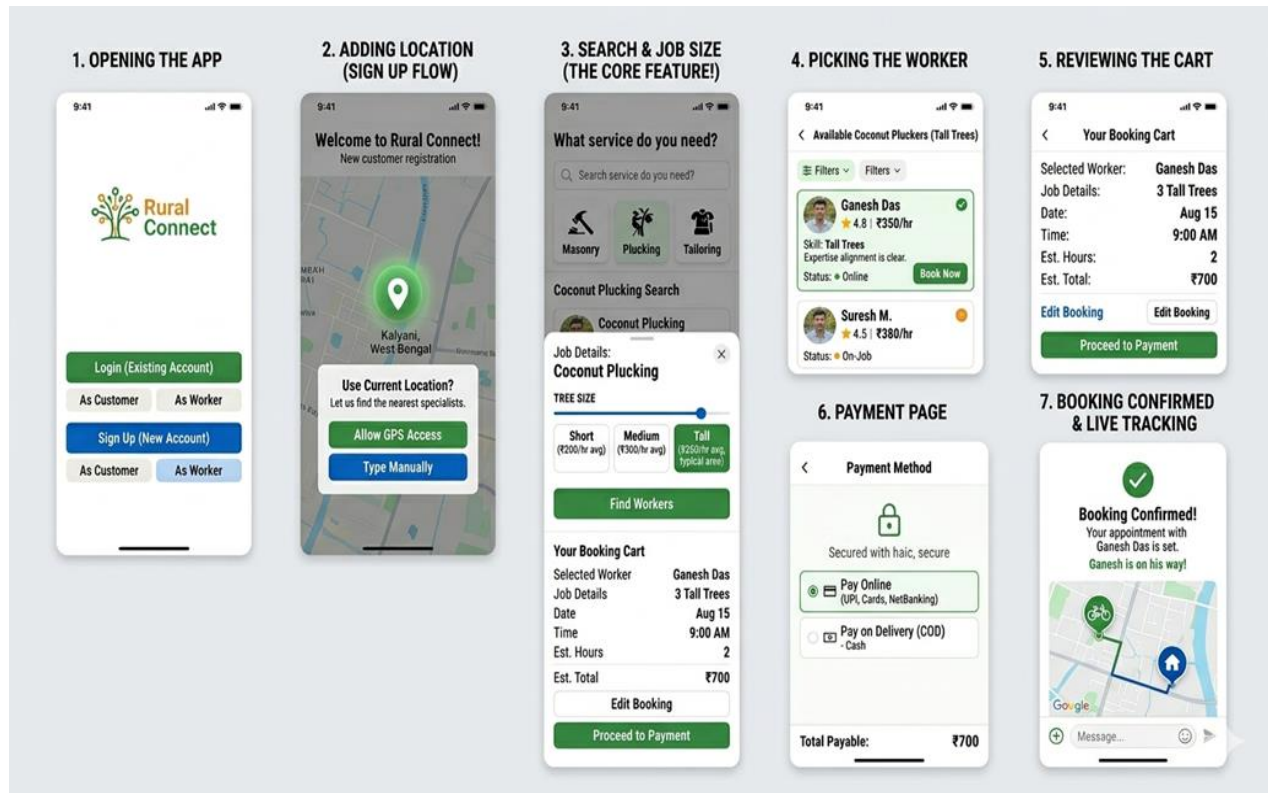
- * FR-3.1: The system must allow customers to manually choose their preferred worker from the roster and push the selection to a confirmation cart.
- * FR-3.2: The checkout module must support multiple payment structures: Digital Pre-payment (UPI, Cards) and Cash-On-Delivery (COD).
- * FR-3.3: For digital pre-payments, the system must hold funds in a secure escrow status, releasing them to the worker account only upon job verification.

2.4 Fulfillment & After-Booking Operations

- * FR-4.1: Upon a confirmed booking, the system must immediately dispatch an app notification to the assigned worker displaying job scope text, height/room tiers, and customer location coordinates.
- * FR-4.2: The system must provide two functional interactive paths for the assigned worker post-booking: Option A (Navigate and Accept) and Option B (Emergency Cancel).
- * FR-4.3: If Option B is selected, the system must flag a critical transaction exception, freeze the worker's status to prevent penalty accumulation, and route the customer back to the alternative worker list.
- * FR-4.4: Upon job completion under Option A, the system must generate a One-Time Password (OTP) on the customer terminal. The worker app must collect and validate this exact OTP to verify job closure and unlock feedback loops.
- * FR-4.5: For COD transactions, the system must block the closing OTP verification until the worker sends a sub-confirmation checking off cash collection.

USER NAVIGATION & OPERATIONS FLOW (APP ARCHITECTURE)

1. The Customer App Navigation Path:



WORKFLOW:

1. Opening the App

- **What happens:** The user opens the app on their phone.
- **The choices:** The app asks them to either **Login** (if they already have an account) or **Sign Up** (if they are new). They must choose whether they are using the app as a **Customer** or a **Worker**.

2. Adding Location

- **What happens:** If it is a new customer signing up, the app asks them to share their **Current Location** using their phone's GPS or by typing it in.
- **Why it matters:** This ensures the app only searches for workers who live close to them.

3. Searching for a Service & Choosing the Size (The Core Feature!)

- **What happens:** The customer types in what they need—for example, "**Coconut Plucking**" or "**Wall Painting**".
- **The Smart Pop-Up:** Before showing any workers, the app shows a simple pop-up asking for the job size:
 - *For Fruit Plucking:* Is the tree Short, Medium, or Tall?
 - *For Painting:* Is it a Single Story or a High-Rise?
- **Showing the Average Price:** Right next to the size choice, the app displays the **Average Local Price** (e.g., "*Tall trees average ₹350/hour in your area*"). This gives the customer an honest price idea before they book.

4. Picking the Worker

- **What happens:** After the customer clicks the size (like *Tall Tree*), a list of local workers appears.
- **What the customer sees:** The customer can look at each worker's profile, including their exact hourly fees, average star ratings out of 5, skills, and if they are currently **Online** or busy **On-Job**.
- **The Choice:** The customer has total freedom to click and pick the exact worker they want based on price or ratings.

5. Reviewing the Cart

- **What happens:** The app takes the customer to a final **Shopping Cart** summary screen.
- **What they double-check:** The customer reviews the booking details to make sure everything is right (e.g., *Selected Worker: Ganesh Das, Job Details: 3 Tall Trees, Estimated Time: 2 Hours*).

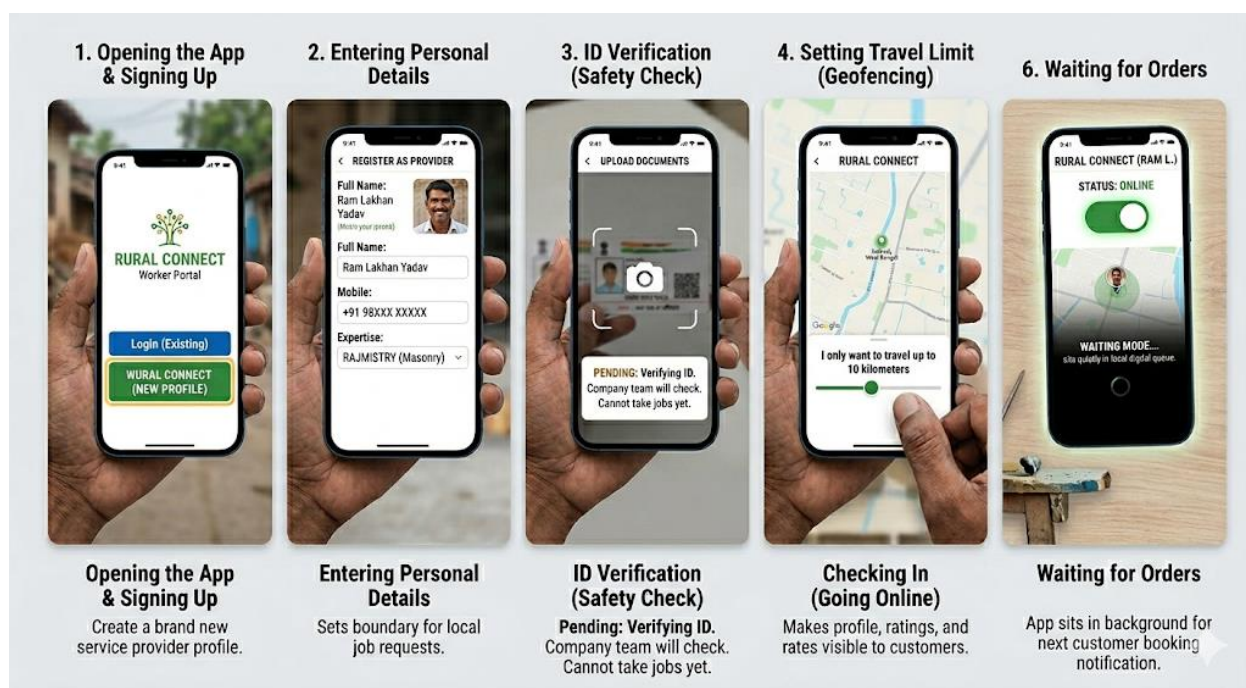
6. Payment Page

- **What happens:** The customer moves to the payment screen.
- **How they pay:** They can pay digitally right away using UPI or Cards, or they can choose **Cash on Delivery (COD)** to pay with physical cash after the job is completely finished.

7. Booking Confirmed & Live Tracking

- **What happens:** Once payment is confirmed (or COD is chosen), the booking is officially saved in the system database.
- **The final screen:** The customer gets a confirmation alert, and the app opens a live map screen where they can track the worker walking or riding to their house.

2. The WORKER App Navigation Path:



WORKFLOW:

1. Opening the App & Signing Up

- **What happens:** The worker opens the app on their phone.
- **The choices:** They select the **Worker Sign-Up** option to create a brand new service provider profile.

2. Entering Personal Details

- **What happens:** The worker types in their basic information.
- **What they enter:** They fill out their **Full Name**, **Mobile Number**, and their specific **Service Expertise** (for example, choosing *Rajmistry*, *AC Service*, or *Fruit Plucking*).

3. ID Verification (Safety Check)

- **What happens:** The worker takes a photo and uploads their government ID card or legal documents.
- **The System Lock:** The app automatically locks their account in a "Pending" status. They cannot take any jobs until the company background team checks and approves their ID for customer safety.

4. Setting the Travel Limit (Geofencing)

- **What happens:** The worker enters their **Location Preference** or sets a maximum distance slider (for example, *"I only want to travel up to 10 kilometers"*).
- **Why it matters:** This sets a boundary so the app only sends them job requests within their preferred area.

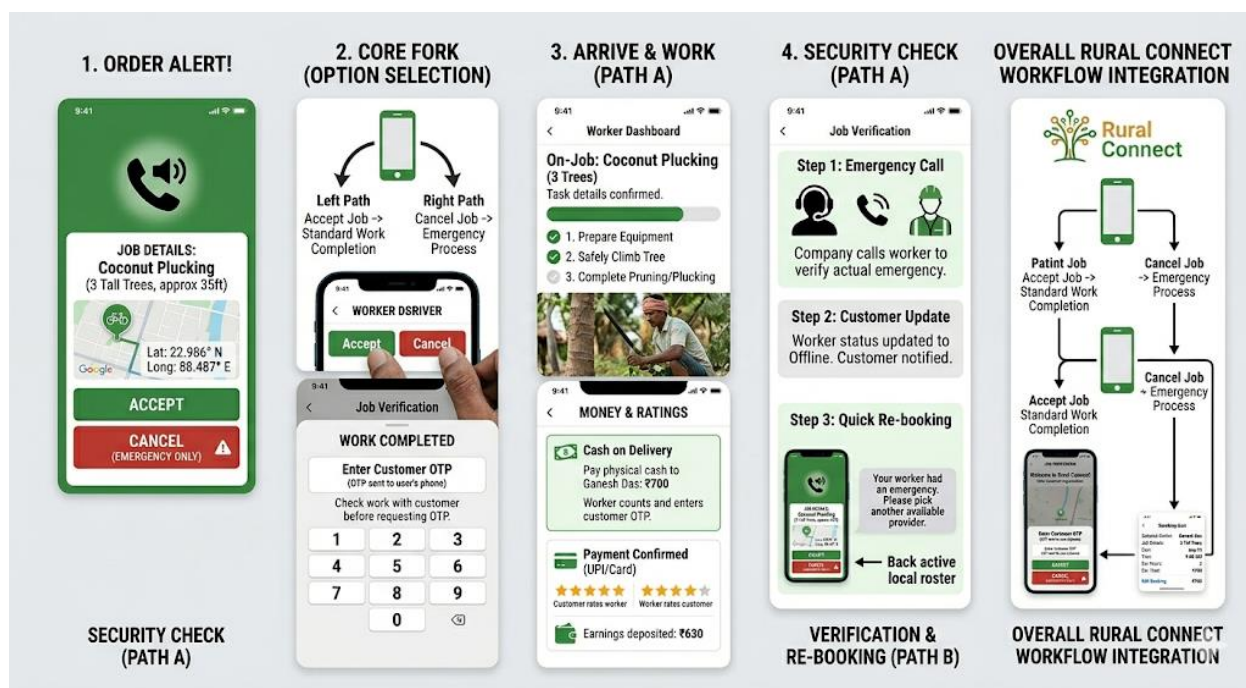
5. Checking In (Going Online)

- **What happens:** Once approved, the worker flips a simple **Check-In** toggle switch inside the app to change their status to **Online**.
- **Why it matters:** This instantly makes them visible on the map, allowing local customers to see their profile, ratings, and rates.

6. Waiting for Orders

- **What happens:** The worker leaves the app open or running in the background.
- **The final state:** The app sits quietly in a waiting mode, placing them in the local digital queue. As soon as a nearby customer books their service tier, a loud order notification will pop up on their screen.

3.The WORKER App Navigation Path:



WORKFLOW:

1. Worker Receives the Order Alert

- **What happens:** The moment a customer confirms a booking, the chosen worker gets a loud notification on their phone.
- **What the worker sees:** The app immediately shows them what the job is, the exact customer location coordinates, and the specific job requirements (for example: "3 Tall Trees, approx 35ft").

2. The Worker's Choice (The Core Fork)

- **What happens:** The worker looks at the screen and must tap one of two options:
 - **Option A:** Accept the job and start navigating to the customer.
 - **Option B:** Cancel the job immediately due to a genuine emergency (like a sudden bike accident on the way), even if they were just online.

🔗 Path A: Standard Work Completion (If they choose Option A)

- **Step 1: Arrive & Work:** The worker clicks navigate, travels to the customer's house, and finishes the assigned task (like pruning the garden or fixing the AC).
- **Step 2: Tap "Work Completed":** Once done, the worker opens their app and clicks the "Work Completed" button.
- **Step 3: The Security OTP Check:** The customer's phone will instantly receive a secret **One-Time Password (OTP)**. The customer only gives this OTP to the worker after checking the work and making sure it is done correctly.
- **Step 4: Handling the Money:**
 - *If paid online already:* The app immediately moves both parties to the Feedback & Rating screen.
 - *If choosing Cash on Delivery (COD):* The customer gets an alert to pay physical cash to the worker. Once the worker counts the cash and enters the customer's OTP, the transaction closes.
- **Step 5: Ratings & Reviews:** Both the customer and worker rate each other out of 5 stars, and the system opens up the worker's digital wallet to deposit their earnings.

⚠️ Path B: Emergency Cancellation (If they choose Option B)

- **Step 1: Company Verification Call:** To prevent workers from canceling lazily, picking Option B flags a major platform exception. The company's customer care team immediately calls the worker to confirm and verify the actual emergency.
- **Step 2: Customer Notification:** Once the team confirms it is a real emergency, the system updates that worker's status to **Offline** (so they don't get penalized) and automatically updates the customer's app.
- **Step 3: Quick Re-booking:** The customer gets a message saying, "Your worker had an emergency. Please pick another available provider." The customer is taken right back to the active local roster to select a new worker, keeping the booking process smooth and continuous.

RISK MANAGEMENT & SPECIAL CASES

This section outlines how the system uses smart software logic to handle real-world problems on the field, protecting both the customer and the worker.

Case A: The Worker No-Show (Fulfillment Failure)

- **The Real-World Risk:** A customer books a *Rajmistry* for urgent wall plastering, but the worker fails to show up, ruining the platform's reputation for reliability.
- **How the App Solves It:**
 - The app tracks the worker's phone location 30 minutes before the job starts. If the worker isn't moving toward the customer, the system flags it.
 - If a worker goes completely silent or cancels last minute, the system's **Emergency Re-routing Engine** kicks in. It takes the exact job details and instantly passes them to the next closest available worker in the same price tier.
 - To keep the customer happy, the app automatically gives them a 10% discount coupon for their next booking, while the original worker receives a penalty score on their profile for breaking reliability.

Case B: Severe Workplace Injury (High-Risk Labor)

- **The Real-World Risk:** An independent worker accidentally falls from a tree during a high-altitude *Coconut Plucking* job or gets hurt while working.
- **How the App Solves It:**
 - To protect everyone from legal and medical stress, the app adds a tiny **Micro-Insurance Fee (₹5 to ₹10)** directly into the customer's total bill.
 - This small fee automatically buys a temporary insurance policy that covers the worker's medical costs for the exact hours they are on that specific job.
 - Additionally, the worker's app features a large **One-Click SOS Emergency Button**. Tapping it instantly broadcasts their exact GPS coordinates to the nearest hospital and alerts the company's emergency team.

Case C: Customer Refuses to Pay or Denies Work Completion (Path A Risk)

- **The Real-World Risk:** A worker finishes fixing an AC or cleaning a bathroom under **Path A**, but the customer lies and says the work isn't done to avoid paying.
- **How the App Solves It:**
 - The platform enforces a strict **Secure OTP Handshake** rule. The worker cannot tap "Job Finished" on their end until they type in a unique 4-digit code generated on the customer's phone.

- The customer is instructed by the app to only hand over this code once they have physically inspected and approved the manual labor.
- For Cash-on-Delivery (COD) jobs, the system double-locks this flow: the worker must check off a "Cash Collected" box before the OTP entry screen unlocks. This creates a digital paper trail that prevents payment fraud and protects the worker's wages.

Case D: En-Route Medical or Personal Crises (Path B Risk)

- **The Real-World Risk:** A worker is genuinely online and wants to do the job, but on the way to the customer's house, they get into a bike accident or face a sudden personal emergency.
- **How the App Solves It:**
 - The worker can tap **Option B ("Cannot Attend Work")** on their active dashboard. This instantly stops the booking timer and freezes the system so the worker doesn't get automatically penalized for a no-show.
 - To prevent workers from abusing this button out of laziness, the system immediately drops this ticket into a high-priority **Admin Verification Queue**. A company customer service agent calls the worker within 5 minutes to verify the emergency.
 - Once confirmed, the system marks the worker as *Offline* so they can rest without hurting their platform rating, and seamlessly sends a notification to the customer saying: *"Your worker had an emergency. Please tap here to select another local provider."*

DATA ARCHITECHTURE

1. ARCHITECTURE OVERVIEW:

To make the "Rural Connect" platform run smoothly, the backend database uses a highly optimized relational design consisting of four core tables:

1. **Service Table:** Stores the master list of all 11 available services categorized into their respective economic sectors.
2. **User Data Table:** Stores customer profiles, locations, and their overall platform trust ratings.
3. **Worker Data Table:** Stores worker profiles, their specific chosen expertise, custom hourly rates, and their real-time check-in status (Online/Offline).
4. **Transaction data Table:** The central log that connects Customers, Workers, and Services together. It records booking dates, hours billed, location, total payments, and the dynamic "Job Requirements" column.

2. DATA ARCHITECTURE DESIGN:

These are the SQL commands to construct the database schema for the project report:

-- 1. Create the Master Service Table

```
CREATE TABLE Service(  
    Catagory_Id VARCHAR(50) PRIMARY KEY,  
    Serv_Name VARCHAR(80) NOT NULL,  
    Sector VARCHAR(80) NOT NULL  
);
```

-- 2. Create the Customer Profile Table

```
CREATE TABLE User_Data(  
    User_Id VARCHAR(50) PRIMARY KEY,  
    User_Name VARCHAR(50) NOT NULL,  
    Email VARCHAR(50) NOT NULL,  
    Mobile_No VARCHAR(50) NOT NULL,  
    Reg_Date VARCHAR(50) NOT NULL,
```

```
Id_Verification VARCHAR(50) DEFAULT('yes'),  
Avg_Rating DECIMAL(3,2)  
);
```

-- 3. Create the Worker Profile Table

```
CREATE TABLE Worker_data(  
    Worker_Id VARCHAR(50) PRIMARY KEY,  
    Name VARCHAR(50) NOT NULL,  
    Expertise VARCHAR(50) NOT NULL,  
    Hourly_Rate VARCHAR(50) NOT NULL,  
    Verification VARCHAR(50) DEFAULT('yes'),  
    CheckIn_Status VARCHAR(50) NOT NULL, -- Online, On-Job, Offline  
    Avg_Rating DECIMAL(3,2)  
);
```

-- 4. Create the central Transaction Table with the flexible requirement layer

```
CREATE TABLE Transaction_data(  
    Booking_Id VARCHAR(50) PRIMARY KEY,  
    Customer_Id VARCHAR(50) NOT NULL,  
    Worker_Id VARCHAR(50) NOT NULL,  
    Booking_Date VARCHAR(50) NOT NULL,  
    Service_Name VARCHAR(50) NOT NULL,  
    Hours_Billed VARCHAR(50) NOT NULL,  
    Total_PaidAmount VARCHAR(50) NOT NULL,  
    Work_Status VARCHAR(60) NOT NULL, -- Completed, Cancelled  
    Job_Requirements VARCHAR(255), -- Stores custom physical job sizes  
    and average rates  
    Location VARCHAR(225)  
);
```

3. DATA INJECTION:

These are the outputs for the tables created:

1. Service Table:

	Catagory_Id	Serv_Name	Sector
1	CAT001	Rajmistry	Civil Construction
2	CAT002	Coconut Plucking	Agriculture
3	CAT003	Mango Harvesting	Agriculture
4	CAT004	Wall Plastering	Civil Construction
5	CAT005	Gardening	Landscaping
6	CAT006	Maid	Domestic Utilities
7	CAT007	Ac service	Appliance Repair
8	CAT008	Fridge service	Appliance Repair

2. User Data Table:

	User_Id	User_Name	Email	Mobile_No	Reg_Date	Id_Verification	Avg_Rating
2	USR002	Priya Nair	priya.nair@mail.com	9876543211	2026-01-15	yes	4.50
3	USR003	Rohan Das	rohan.das@mail.com	9876543212	2026-02-01	yes	4.90
4	USR004	Sneha Ch...	sneha.c@mail.com	9876543213	2026-02-18	yes	4.20
5	USR005	Vikram Ma...	vikram.m@mail.com	9876543214	2026-03-05	yes	4.60
6	USR006	Ananya R...	ananya.r@mail.com	9876543215	2026-03-22	yes	4.75
7	USR007	Abhishek ...	abhishek.j@mail.com	9876543216	2026-04-02	yes	4.40
8	USR008	Meera Ka...	meera.k@mail.com	9876543217	2026-04-11	yes	4.95
9	USR009	Sandeep ...	sandeep.v@mail.com	9876543218	2026-05-01	yes	3.80

3. Worker Data Table:

	Worker_Id	Name	Expertise	Hourly_Rate	Verification	CheckIn_Status	Avg_Rating
2	WKR002	Suresh Mondal	Rajmistry	180	yes	Online	4.20
3	WKR003	Ganesh Das	Coconut Plucking	300	yes	On-Job	4.90
4	WKR004	Vikram Singh	Mango Harvesting	150	yes	Online	3.50
5	WKR005	Binod Paswan	Rajmistry	220	yes	Offline	4.40
6	WKR006	Laltu Sheikh	Coconut Plucking	280	yes	Online	4.65
7	WKR007	Madan Mumu	Mango Harvesting	160	yes	Offline	4.10
8	WKR008	Hari Charan	Wall Plastering	200	yes	Online	4.50
9	WKR009	Sukumar Roy	Wall Plastering	240	yes	On-Job	4.80
10	WKR010	Arjun Thapa	Gardening	140	yes	Online	4.30
11	WKR011	Laxmi Bai	Maid	100	yes	Online	4.60
12	WKR012	Saraswati Das	Maid	90	yes	Offline	4.15
13	WKR013	Rajesh Yadav	Ac service	350	yes	Online	4.70
14	WKR014	Nitin Chawla	Ac service	400	yes	On-Job	4.85
15	WKR015	Somnath Paul	Fridge service	300	yes	Online	4.40
16	WKR016	Karan Johar	Bathroom cleaning	150	yes	Online	4.25
17	WKR017	Manoj Tiwari	Bathroom cleaning	130	yes	Offline	3.90
18	WKR018	Subhash Dutta	Decoration(Stage)	500	yes	Online	4.95
19	WKR019	Joydeep Pal	Decoration(Light...	350	yes	Online	4.75
20	WKR020	Aniket Shaw	Gardening	160	yes	Offline	4.50
21	WKR021	Sanjay Dutt	Fridge service	320	yes	Online	4.20
22	WKR022	Sanjoy Dutta	Rajmistry	260	yes	Online	4.95

4. Transaction Data Table:

	Booking_Id	Customer_Id	Worker_Id	Booking_Date	Service_Name	Hours_Billed	Total_PaidAmount	Work_Status	Job_Requirements	Location
1	BKG001	USR001	WKR001	2026-01-12	Rajmistry	4	1000	Completed	NULL	Garia
2	BKG002	USR002	WKR002	2026-01-16	Rajmistry	5	900	Completed	NULL	Garia
3	BKG003	USR003	WKR003	2026-02-02	Coconut Plucking	2	600	Completed	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr Equip...	Garia
4	BKG004	USR004	WKR004	2026-02-20	Mango Harvesting	6	900	Cancelled	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr Equip...	Garia
5	BKG005	USR005	WKR005	2026-03-06	Rajmistry	8	1760	Completed	NULL	Garia
6	BKG006	USR006	WKR006	2026-03-24	Coconut Plucking	3	840	Completed	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr Equip...	Garia
7	BKG007	USR007	WKR007	2026-04-03	Mango Harvesting	4	640	Completed	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr Equip...	Garia
8	BKG008	USR008	WKR008	2026-04-12	Wall Plastering	5	1000	Completed	Tier 2: Double Story Exterior Avg Rate: 240/hr Sc...	Garia
9	BKG009	USR009	WKR009	2026-05-02	Wall Plastering	4	960	Worker No-Show	Tier 2: Double Story Exterior Avg Rate: 240/hr Sc...	Garia
10	BKG010	USR010	WKR010	2026-05-14	Gardening	3	420	Completed	NULL	Garia
11	BKG011	USR011	WKR011	2026-05-19	Maid	4	400	Completed	NULL	Garia
12	BKG012	USR012	WKR013	2026-05-21	Ac service	2	700	Completed	NULL	Garia
13	BKG013	USR001	WKR015	2026-01-15	Fridge service	3	900	Completed	NULL	Garia
14	BKG014	USR002	WKR016	2026-01-20	Bathroom cleaning	2	300	Completed	NULL	Garia
15	BKG015	USR003	WKR018	2026-02-05	Decoration(Stage)	10	5000	Completed	Tier 2: Medium Scope (3 Rooms + Balcony) Avg R...	Garia
16	BKG016	USR004	WKR019	2026-02-22	Decoration(Lighting)	8	2800	Completed	Tier 2: Medium Scope (3 Rooms + Balcony) Avg R...	Dum Dum
17	BKG017	USR005	WKR020	2026-03-10	Gardening	5	800	Completed	NULL	Dum Dum
18	BKG018	USR006	WKR021	2026-03-28	Fridge service	2	640	Cancelled	NULL	Dum Dum
19	BKG019	USR007	WKR022	2026-04-05	Rajmistry	6	1560	Completed	NULL	Dum Dum
20	BKG020	USR008	WKR012	2026-04-15	Maid	6	540	Completed	NULL	Dum Dum
21	BKG021	USR009	WKR014	2026-05-04	Ac service	3	1200	Completed	NULL	Dum Dum
22	BKG022	USR010	WKR017	2026-05-16	Bathroom cleaning	4	620	Completed	NULL	Dum Dum

5. How Sql tables work through different filters operation in advance level:

Problem 1: The High-Risk Tier Filter

Business Scenario: Suppose us wants to check how many high-altitude jobs are being booked to evaluate insurance liabilities and worker safety compliance.

SQL QUERY:

```
select Booking_Date,Booking_Id,Service_Name,Job_Requirements from Transaction_data
where Job_Requirements='Tier 3: Tall Tree (>30ft) | Avg Rate: 350/hr | Equipped: Yes';
```

	Booking_Date	Booking_Id	Service_Name	Job_Requirements
1	2026-02-02	BKG003	Coconut Plucking	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
2	2026-02-20	BKG004	Mango Harvesting	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
3	2026-03-24	BKG006	Coconut Plucking	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
4	2026-04-03	BKG007	Mango Harvesting	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
5	2026-05-28	BKG024	Coconut Plucking	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
6	2026-05-15	BKG033	Mango Harvesting	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
7	2026-05-20	BKG034	Coconut Plucking	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
8	2026-04-12	BKG042	Coconut Plucking	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
9	2026-04-20	BKG043	Mango Harvesting	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...
10	2026-05-08	BKG045	Mango Harvesting	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr E...

Problem 2: The Online and Affordable Roster

Business Scenario: To assist customer support in matching immediate user requests with cost-effective options, this query identifies available workers who don't charge premium rates.

SQL QUERY:

```
select Name,Expertise,Hourly_Rate from Worker_data
where CheckIn_Status='online'and Hourly_Rate<=300;
```

	Name	Expertise	Hourly_Rate
1	Ramesh Kumar	Rajmistry	250
2	Suresh Mondal	Rajmistry	180
3	Vikram Singh	Mango Harvesting	150
4	Laltu Sheikh	Coconut Plucking	280
5	Hari Charan	Wall Plastering	200
6	Arjun Thapa	Gardening	140
7	Laxmi Bai	Maid	100
8	Somnath Paul	Fridge service	300
9	Karan Johar	Bathroom cleaning	150
10	Sanjoy Dutta	Rajmistry	260

Problem 3: The High-Value Transaction Tracker

Business Scenario: You need to audit large cash flows to calculate platform commission from premium services.

SQL QUERY:

```
select * from Transaction_data
where Work_Status='Completed' and Total_PaidAmount>1000;
```

	Booking_Id	Customer_Id	Worker_Id	Booking_Date	Service_Name	Hours_Billed	Total_PaidAmount	Work_Status	Job_Requirements	Location
1	BKG005	USR005	WKR005	2026-03-06	Rajmistry	8	1760	Completed	NULL	Garia
2	BKG015	USR003	WKR018	2026-02-05	Decoration(Stage)	10	5000	Completed	Tier 2: Medium Scope (3 Rooms + Balcony) Avg R...	Garia
3	BKG016	USR004	WKR019	2026-02-22	Decoration(Lighting)	8	2800	Completed	Tier 2: Medium Scope (3 Rooms + Balcony) Avg R...	Dum Dum
4	BKG019	USR007	WKR022	2026-04-05	Rajmistry	6	1560	Completed	NULL	Dum Dum
5	BKG021	USR009	WKR014	2026-05-04	Ac service	3	1200	Completed	NULL	Dum Dum
6	BKG023	USR011	WKR001	2026-05-25	Rajmistry	5	1250	Completed	NULL	Dum Dum
7	BKG028	USR004	WKR018	2026-03-15	Decoration(Stage)	6	3000	Completed	Tier 2: Medium Scope (3 Rooms + Balcony) Avg R...	Dum Dum
8	BKG029	USR005	WKR019	2026-04-02	Decoration(Lighting)	5	1750	Completed	Tier 2: Medium Scope (3 Rooms + Balcony) Avg R...	Dum Dum
9	BKG032	USR008	WKR002	2026-05-10	Rajmistry	7	1260	Completed	NULL	Salt Lake Sector 5
10	BKG033	USR009	WKR004	2026-05-15	Mango Harvesting	10	1500	Completed	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr Equipp...	Salt Lake Sector 5
11	BKG034	USR010	WKR006	2026-05-20	Coconut Plucking	4	1120	Completed	Tier 3: Tall Tree (>30ft) Avg Rate: 350/hr Equipp...	Salt Lake Sector 5
12	BKG035	USR011	WKR008	2026-05-22	Wall Plastering	6	1200	Completed	Tier 2: Double Story Exterior Avg Rate: 240/hr Sc...	Salt Lake Sector 5
13	BKG037	USR001	WKR014	2026-03-01	Ac service	3	1200	Completed	NULL	Salt Lake Sector 5
14	BKG038	USR002	WKR015	2026-03-05	Fridge service	4	1200	Completed	NULL	Salt Lake Sector 5
15	BKG041	USR005	WKR001	2026-04-05	Rajmistry	5	1250	Completed	NULL	Salt Lake Sector 5
16	BKG044	USR008	WKR005	2026-05-02	Rajmistry	6	1320	Completed	NULL	Salt Lake Sector 5
17	BKG046	USR010	WKR009	2026-05-14	Wall Plastering	5	1200	Completed	Tier 2: Double Story Exterior Avg Rate: 240/hr Sc...	New Town
18	BKG048	USR012	WKR013	2026-05-29	Ac service	4	1400	Completed	NULL	New Town

Rural Connect Operational Performance Dashboard



Objective: Developed an interactive, data-driven Power BI dashboard for "Rural Connect"—a platform connecting local service professionals (such as construction workers, appliance technicians, and specialized harvesting teams) with regional consumer demand. The primary business objective was to bridge operational gaps, optimize service workflows, track workforce performance, and map out hyper-local demand trends across 7 distinct regions in Kolkata.

Key Dashboard Components & Analytics Breakdown:

- **Executive KPI Cards (Top Banner):** Tracks high-level ecosystem health at a glance by displaying total operational transactions (43.67K), absolute unique bookings (230), cumulative platform sales (409K), and active onboarded service workers (22).
- **Operational Health & Efficiency Trackers (Left Corridor):**
 - * **Work Status Donut Chart:** Monitors contract conversion cycles (Completed vs. Cancelled).
 - **Tactical Performance Cards:** Displays dynamic operational efficiency metrics, showing an average fulfillment speed of **6 Hours per booking** and a high platform **Fulfillment Rate of 97%** (calculated using conditional row-level DAX aggregation).
- **Macro Trend Tracking (Center Top):** A line chart tracking booking volumes by month, showing a clear growth trajectory from baseline numbers in January scaling up to a peak of 125 bookings by June.

- **Service & Worker Performance Matrix (Center Bottom):** Integrates a dynamic bar chart highlighting top-grossing services (led heavily by *Rajmistry* at 83K and *Maid* services at 55K) alongside a granular Worker Performance Grid showing individual IDs, average ratings, and direct revenue generated.
- **Market Mechanics (Right Corridor):**
 - * **Rating vs. Pricing Elasticity Scatter Plot:** Analyzes if higher hourly rates impact customer ratings, providing pricing optimization insights for service categories.
 - **Geographic Treemap:** Maps regional consumer density dynamically across 7 target zones (Singur, Behala, Bonhooghly, Dum Dum, New Town, Salt Lake Sector 5, and Garia), revealing structural demands like heavy agricultural harvesting spikes in Singur vs. core appliance maintenance inside urban hubs.

Conclusion & Future Scope

Conclusion:

The "Rural Connect" project successfully demonstrates how raw, fragmented transactional data can be transformed into a centralized, highly strategic business intelligence asset. By leveraging SQL Server for robust data modeling and constraint management, alongside Power BI for dynamic visualization and custom DAX metrics, the project has provided a 360-degree view of the platform's ecosystem health.

Key milestones achieved through this analytics framework include:

- **Operational Transparency:** Instant visibility into vital business health markers, establishing that the platform operates at a high **97% Fulfillment Rate** with an efficient **6-hour average turnaround time**.
- **Socio-Economic Insights:** The granular mapping of demand across 7 distinct zones in Kolkata highlighted the structural differences between rural areas (e.g., specialized harvesting in Singur) and urban tech corridors (e.g., appliance maintenance in Salt Lake Sector 5).
- **Workforce Empowerment:** The integration of rating and pricing elasticity metrics gives stakeholders the tools needed to optimize service pricing while maintaining high quality standards.

Ultimately, this dashboard transitions "Rural Connect" from a reactive operational model to a proactive, data-driven strategy, enabling data-backed decisions that drive platform growth, elevate user experiences, and support local livelihoods.

Future Scope:

To maintain a competitive edge and build upon the foundational analytics established in this project, the next phase of "Rural Connect" will focus on scaling platform capabilities, enhancing customer convenience, and optimizing worker utility through three major strategic pillars:

☐ **1. AI-Powered Chatbot Integration & Smart Dispatch**

The integration of a conversational AI chatbot will revolutionize how users interact with the platform. Rather than manually navigating filters, customers can simply type or speak their requirements in plain or regional languages (e.g., *"I need an AC technician tonight"*).

- **Impact on Analytics:** The AI will parse unstructured text into structured database fields instantly. It will automatically categorize job complexities, predict job duration, and dynamically route orders to the highest-rated, closest available "Online" workers, driving the platform fulfillment rate closer to 100%.

2. Predictive Advance Booking System

Transitioning from an immediate, on-demand dispatch model to an advanced scheduling roster will unlock predictable revenue streams. Customers will be able to book service professionals days, weeks, or months in advance—perfect for seasonal agricultural harvesting cycles in fringe areas or routine monthly residential maintenance.

- **Impact on Analytics:** This will allow the Power BI dashboard to transition from displaying purely historical trends to offering **predictive capacity planning**. Operations teams will be able to forecast labor shortages or demand spikes by location weeks before they happen, allowing for targeted supply recruitment.

3. Multi-Work Bundling & Service Combos

To maximize the average transaction value (ATV) and provide a seamless user experience, the platform will introduce a multi-work combo facility. Customers will no longer need to book separate appointments for different household chores; instead, they can book cohesive bundles (e.g., an *AC Service* + *Deep Bathroom Cleaning* combo).

- **Impact on Analytics:** For workers, this introduces multi-skill utilization, allowing a single versatile professional to maximize their earnings per house visit. For the analytical pipeline, this introduces cross-selling data structures, allowing the business to analyze which service pairs have the highest affinity, optimizing marketing campaigns and bundle pricing strategies.